Skin Lesion Classification with Explainability

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ABSTRACT

Accessible skin lesion classification tools remain a challenge to create and confidently deploy in medical imaging due to the lack of publicly available balanced datasets and the variety in skin tones. This paper presents a deep-learning framework for the classification of skin lesions, using the EfficientNet-B4 architecture and a custom dataset. To address the class imbalance in the HAM10000 dataset, we integrated the ISIC 2019 dataset's malignant class and implemented data pipelines such as the Albumentations-based data augmentation, as well as weighted random sampling. Our methodology expands user interpretability and collaboration by developing an explainability dashboard to visualize model decision-making through Grad-CAM and NLP, which allows for improved user examination. Experimental results demonstrate that the proposed model achieves a test accuracy of 96.62%, an AU-ROC Score of 0.9946, an F1 score of 0.9707, and a recall of 0.9642. Additionally, the model achieves sensitivity at a fixed capacity of 0.9893 at 10% false positive rate. These results suggest that Convolutional Neural Networks, paired with robust data handling techniques, can provide high-precision and accurate diagnostic tools for dermatological screening of skin lesions.

CCS Concepts

- Computing methodologies → Modeling
- Computing methodologies → Machine Learning
- Computing methodologies → Artificial Intelligence
- Computing methodologies → Computer Vision
- Applied Computing → Healthcare

Keywords

Machine Learning, Artificial Intelligence, Convolutional Neural Networks, CNN, Deep Learning, EfficientNet, Explainable AI, Skin Lesions, Computer Vision, Healthcare, Skin Lesion Classification, Binary Classification, Transfer Learning

1. Introduction and Background

1.1 Background

Areas of skin differing from nearby tissue define what are known as skin lesions. Resulting sometimes from trauma or external harm, these changes appear for various reasons. When a skin lesion turns harmful, it gains the ability to expand without restraint, moving into adjacent regions of the body. As noted by Shetty et al. [12], categorization splits them into two forms: primary and secondary. Present at birth or emerging slowly, primary ones reflect original abnormalities in the skin's structure. Unlike primary ones, secondary lesions arise when initial abnormalities worsen or transform. A scab resulting from scratching a mole illustrates such a change. Depending on the case, management might involve self-care, drugs, or surgical removal. When cancerous, as with melanoma, these areas pose a serious risk due to their ability to infiltrate local structures and migrate internally, often life-threatening without timely recognition. Correct identification becomes essential. Therefore, spotting malignant traits fast allows a proper response, supporting medical safety [12].

There is a noticeable gap in accessible and immediate care (tools or consultations) in addressing skin complications. Dermatology training in various medical schools has proven to be inadequate or non-existent in addressing the increasing demand for skin consultations, resulting in waiting times of up to 12 months in high-income countries, while those in medium- or low-income countries may never gain assistance [7]. This issue was brought upon and heavily emphasized in an event held during the 77th World Health Assembly, where it was noted that skin diseases ranked as one of the most common reasons why people seek assistance from healthcare providers [7]. The emergence of machine learning technology (ML) and artificial intelligence (AI) has opened a gateway to building solutions for specialized problems, with healthcare proving to be a popular selection for ML-centric solutions. Healthcare fields such as radiology, genetics, and neuroimaging have incorporated these tools to improve the accuracy of diagnosis and the overall outcome of patients [9]. Dermatology in particular has proven to be a suitable field for AI and ML integration due to its reliance on images and visual data.

A study conducted by Chen [3] shows that expert and non-expert human classifiers show strong performance in classifying malignant skin lesions, with the expert achieving an accuracy of 93.59% and the non-expert achieving 94.87%. However, both human classifiers showed significantly lower performance in identifying benign lesions, with the expert scoring an accuracy of 32.86% and the non-expert scoring 28.64%. In comparison, the deep-learning model achieved more consistent and stable performance in identifying both malignant and benign skin lesions, with an accuracy of 93.59% for malignant cases and 70.42% for benign lesions. These findings indicate that while human evaluators are effective at detecting malignant lesions, their ability to accurately identify benign lesions remains limited, highlighting the potential of artificial intelligence-based systems to improve diagnostic consistency and reduce misclassification in skin lesion analysis.

To bridge the gap between high-performance classification and clinical interpretability, this study proposes the development of an automated, explainable binary classification model for skin lesions. Leveraging Convolutional Neural Network (CNN) architectures specifically utilizing transfer learning with EfficientNetv4, the proposed system aims to accurately distinguish between benign and malignant cases. Furthermore, to ensure diagnostic transparency and technical clarity, the model will incorporate Gradient-weighted Class Activation Mapping (Grad-CAM) to validate its focus on skin features and characteristics.

1.2 Problem Statement

Despite the increasing adoption of artificial intelligence in healthcare, accessible and reliable tools for skin lesion classification remain limited due to challenges such as dataset imbalance, variability in lesion appearance, inconsistency in imaging conditions, and lack of interpretability or explainability in model predictions, which limits trust and reduces usability in real-world diagnostic settings. Furthermore, the skepticism among dermatologists persists due to the absence of transparent reasoning behind model predictions, emphasizing the need for explainable systems that justify or prove their outputs. This highlights the pressing need to develop a robust and explainable skin lesion classification system that not only achieves high diagnostic performance but also provides interpretable and justified outputs across real-world conditions.

1.3 Research Objectives

This study aims to develop a robust and explainable deep learning-based system for the classification of benign and malignant skin lesions. Specifically, the objectives of this study are to: (1) design and implement a convolutional neural network (CNN) model capable of accurately distinguishing between benign and malignant skin lesions; (2) improve model generalization and robustness by addressing common dataset challenges such as class imbalance, lesion variations, and inconsistencies in image quality through data augmentation, specifically Albumentation, and other optimized training strategies; (3) evaluate the performance of the proposed classification model using standard statistical metrics, specifically the precision, recall, f1-score, and receiver operating characteristic curve (AU-ROC), to ensure and validate the reliability and consistency of the model cross different data distributions and real-world scenarios; (4) integrate explainability methods such as Gradient-weighted Class Activation Mapping (Grad-CAM), to provide visual interpretation and justification of model predictions; and (5) integrate uncertainty estimation using Monte Carlo (MC) Dropout to calculate prediction confidence to improve reliability of the model, also promoting transparency and trust among users. Ultimately, this study seeks to contribute to the development and advancement of accurate, interpretable, and explainable tools that can improve decision-making and trust gathering in dermatology.

1.4 Contributions

This research delivers several pivotal advancements in automated dermatological screening, effectively bridging the divide between theoretical machine learning performance and dependable clinical implementation. The core contributions of this project encompass the following:

First, the research provides a crucial breakthrough in dataset engineering by systematically neutralizing the statistical biases that commonly plague open-source medical image repositories. By purposefully extracting underrepresented malignant cases from the ISIC 2019 Challenge dataset and fusing them with the heavily benign-skewed HAM10000 collection, the resulting architecture actively suppresses the algorithm's algorithmic inclination to guess non-cancerous outcomes by default. This structural parity elevates patient safety above superficial accuracy metrics by drastically curtailing the likelihood of life-threatening false negatives.

Second, the study constructs a highly durable image preprocessing pipeline designed specifically to withstand the visual unpredictability of real-world clinical environments. Leveraging the Albumentations library, the training protocol introduces complex optical warping, grid distortions, and synthetic visual interference, such as color jittering and Gaussian blurs, to artificially replicate the physical flaws of amateur, user-captured photography. This rigorous augmentation strategy forces the EfficientNet-B4 framework to bypass shallow digital artifacts and strictly evaluate core morphological lesion traits, thereby maximizing diagnostic consistency across varying illumination and camera angles.

Third, to dismantle the opaque "black-box" nature of traditional deep learning, this study pioneers a dual-layered interpretability framework that fosters physician trust. By synergizing Gradient-weighted Class Activation Mapping (Grad-CAM) to spatially highlight critical lesion characteristics with Monte Carlo (MC) Dropout to mathematically quantify prediction confidence, the classifier transforms into an interactive decision-support mechanism. This transparency ensures that medical professionals can scrutinize the network's visual rationale and proactively gauge its diagnostic certainty prior to finalizing clinical assessments.

Finally, the research introduces an innovative, multi-modal Personalized Safety Engine that merges computer vision with natural language processing. Utilizing a DistilBERT-powered analysis of clinical text alongside a Reinforcement Learning agent, the system's dashboard adaptively recalibrates its diagnostic sensitivity thresholds in response to patient metadata and textual risk indicators. This context-sensitive fusion guarantees that visual diagnostic confidence can be prudently overridden if high-risk textual markers are identified, establishing a highly rigorous, fail-safe clinical evaluation tool.

2. Review of Related Literature

A significant limitation in automated skin lesion classification systems is the underrepresentation and handling of darker skin tones in dermatological image datasets, which may lead to biased and skewed model performance. According to Shabu [11], many widely used datasets contain a disproportionately large number of lighter skin tone images compared to darker skin types, resulting in significant skin tone

imbalance. For instance, the International Skin Imaging Collaboration (ISIC) dataset, which contains 17,728 images, shows that approximately 70% depict light skin tones while only 8% represent dark skin tones. Such an imbalance may cause machine learning models to perform significantly worse when diagnosing lesions on darker skin tones. To address the lack of diversity in dermatological datasets, Daneshjou et al. [5] developed the Diverse Dermatology Images [DDI] dataset, which contains 656 clinically verified dermatology images from 570 patients across Fitzpatrick skin types I-VI. The dataset was designed to evaluate dermatology model systems on a more diverse range of skin tones. From the results and findings of the study upon using the dataset, the authors found that the dermatology AI models experienced a decrease in ROC-AUC performance when evaluated on diverse skin tones compared to their originally reported results, dropping from ROC-AUC values of 0.88-0.92 to 0.65-0.67. The study also reported that both the human evaluators and machine learning models demonstrated low performance or accuracy when evaluating darker skin tones. These findings highlight the importance of developing dermatological machine learning models trained on more diverse datasets to ensure reliable and consistent diagnostic performance across different skin tones.

A study by Lariba and Akanwanzon [10] shows significant progress in developing computer-aided systems for skin cancer classification and diagnosis. Early methods mainly relied on statistical analysis and multi-step detection techniques. For instance, Fatima et al. introduced a computer-assisted six-step approach that analyzes set lesion parameters to detect melanoma. Qian et al. proposed a two-stage framework that first identifies the lesion location before classification, showing better results than single-stage methods. Machine learning techniques were also examined. Ercal et al. utilized artificial neural networks to distinguish between malignant melanoma and benign tumors. They recommended using fuzzy logic and hierarchical neural networks to improve classification performance. When evaluating the performance of automated skin lesion detection, reducing computational overhead without sacrificing accuracy remains a primary challenge. Notably, the authors demonstrated that their custom isolated model required significantly less training time compared to heavy, traditional pretrained models like ResNet50 and DenseNet-201, establishing a strong benchmark for resource-efficient diagnostic tools.

Ali et al. [2] have demonstrated the effectiveness and performance of EfficientNet architectures, particularly B0 up to B7 variants, in the classification of multiclass skin cancer using the HAM10000 dataset. The researchers utilized transfer learning with ImageNet, then further fine-tuned the model to categorize the images into various lesion classes. The results indicated that EfficientNet-B4 and B5 proved to be the most effective variants among the other variants, with B4 achieving the highest accuracy of 87.91% and an F1-Score of 87%. While the lowest performing variant is the B0, with a 5% difference in recall and F1 scores from the B4. Additionally, the study has provided room to further increase the variability and reliability of the model by acknowledging the limitations of using a single dataset. This allows future researchers to incorporate other datasets for variability, as the model utilized only a single dataset (HAM10000), which could lead to the model overfitting.

3. Dataset and Preprocessing

3.1 Data Collection

The custom dataset that we used to develop the binary skin lesion classification model is made up of two open-source datasets. The main dataset is the HAM10000, while the ISIC 2019 is the secondary dataset. Both datasets are provided under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license, which permits the sharing, modification, and utilization of the dataset for non-commercial, academic, and research purposes [6]. We used ISIC 2019 to add data in order to address the class imbalance of the main dataset by using only the malignant classes. The image data from the main and secondary datasets have all undergone prior data cleaning and preprocessing techniques, with the files publicly available on Kaggle having no missing or duplicate data. This process may have included removing files based on image quality, file formatting, and the lack of important visual features in the images. However, to add confidence that the data we're using isn't missing or corrupted, we also added directory or file checks throughout the code to make sure that the dataset's subsets (e.g, train, val, and test) are not empty. Both datasets are downloaded through Kaggle's API and are stored in Google Colab's cloud storage.

3.2 Data Characteristics

The custom dataset has two classes, with the malignant class consisting of melanoma, basal cell carcinoma, squamous cell carcinoma, and actinic keratosis, all encoded as 1. Meanwhile, the benign class consists of nevus, benign keratosis-like lesions, vascular skin lesions, and dermatofibroma, all encoded as 0. This brings the total number of images in the custom dataset to 19,335. The custom dataset that we made is split into three parts, 80% training, 10% validation, and 10% testing. To make sure that the ratio of the subsets remains the same as the original, we also applied stratification. This creates subsets that are similar to the original dataset's ratio of malignant and benign classes. Reproducibility is maintained by applying a fixed random seed (42), which ensures that the ratio of data in the new subsets will always be the same during data pipeline execution.

The custom dataset represents a wide variety of morphological features, ranging from physical shapes, colors, and the pattern of the skin lesions. This makes it difficult for traditional visual-based diagnostic systems to detect subtle malignant lesions, as malignant melanoma may look completely different from one patient to another. Conversely, this also applies to benign skin lesions that bear resemblance to malignant ones. Additionally, the images also contain significant non-morphological features such as dense body hair, reflection from wet or oily skin, streaks from image capture, and clinical marker outlines. The subtleness of the visual features often leads to false classifications within traditional diagnostic systems. Due to the subtlety of the established visual features, EfficientNet-B4 was utilized for its deep architecture and its inherent design to learn micro-patterns that simpler models might miss.

3.3 Data Augmentation

All images undergo preprocessing techniques, namely resizing, color space conversion, transformations, and normalization. Images are resized to the EfficientNet-B4 standard of 224x224, and the color space is converted from BGR to RGB using cv2.cvtColor. Image transformation includes conversion from HWC (OpenCV standard) to the CHW format required for PyTorch, which also helps optimize the training process since it's lighter, as we used Google Colab's free tier for model training [9]. Images also undergo universal standard normalization for ImageNet pre-trained models (e.g, EfficientNet-B4) with a mean of 0.485, 0.456, and 0.406, and standard deviation (std) of 0.229, 0.224, and 0.225. These values were specified on the official website of Albumentations, which is why we used them [1].

The data augmentation pipeline is only applied to the train subset and also uses Python's Albumentations library [8]. For the initial training,

we used the usual augmentation pipeline (e.g, basic flips, color change) libraries, which produced lower metrics since they aren't as robust. The new pipeline includes spatial augmentations, the same horizontal and vertical flips, but added random rotation of 90 degrees, all of which have a probability of 0.5. Advanced distortion techniques, such as grid and optical distortion, were also applied to the subset, which mimics skin elasticity (reflection/skin) and changing angles in real-life image capture. Variety in lighting and coloring was also simulated through color jitter, which changed the image's brightness, contrast, saturation, and hue. To simulate variety in image capture quality and the usual image obstructions, Gaussian noise was applied to ensure that the model learns to look past shallow obstructions and learn harder to identify visual features such as texture and color patterns.

4. Methodology

4.1 Model Architecture

The system utilizes the EfficientNet-B4 architecture, selected for its efficient compound scaling of depth, width, and resolution. This architecture is particularly adept at dermatological tasks because its high-resolution input facilitates the identification of fine-grained micro-patterns and irregular pigment networks often missed by shallower models. The project employs transfer learning using weights pre-trained on ImageNet to adapt the model for binary classification. The final fully connected layer was modified to output two specific classes: benign (encoded as 0) and malignant (encoded as 1). This deep architecture provides a high-dimensional feature extractor capable of characterizing complex lesion textures and borders across varied imaging environments.

4.2 Implementation Details

The core of our diagnostic framework relies on the EfficientNet-B4 model, favored for its balanced compound scaling and capacity to detect intricate dermatological details. By initiating the network with pre-trained ImageNet parameters, we capitalized on existing visual feature recognition capabilities, while tailoring the final fully connected layer to yield a binary output mapping benign lesions to 0 and malignant ones to 1. Early on, the profound class imbalance within the primary HAM10000 dataset posed a hurdle, pushing the model to generate overly safe, non-cancerous predictions. To resolve this, we integrated a WeightedRandomSampler that adjusted sampling probabilities based on class distribution, guaranteeing an equitable 50/50 ratio of malignant to benign examples within every training batch.

Throughout the learning phase, optimization was guided by a weighted CrossEntropyLoss metric. By taking the inverse square root of the sample counts per class, this function applied a severe mathematical penalty for misidentifying malignant cases, ensuring that the network prioritized these critical instances during backpropagation. We employed the AdamW optimizer, setting a learning rate of 0.0002 and a weight decay of 0.01, specifically to take advantage of its decoupled weight regularization. Additionally, a ReduceLROnPlateau scheduler actively tracked validation loss. If the loss stagnated for three consecutive epochs, the scheduler scaled down the learning rate by a factor of 0.1, facilitating more granular parameter updates.

To ensure the model generalized well to real-world scenarios, we constructed an extensive preprocessing and augmentation workflow utilizing the Albumentations library. This setup introduced spatial alterations, including random 90-degree rotations as well as vertical and horizontal flips, mimicking diverse camera orientations. Furthermore, optical and grid distortions were utilized to replicate the effects of skin stretching and lens curvature. By injecting Gaussian noise and color jitter, the pipeline also simulated varying illumination, sensor artifacts, and low-quality user photographs, compelling the network to focus on essential morphological traits rather than superficial image defects.

While the baseline training process was configured for 30 epochs, strict runtime limitations of 90 minutes required capping the standard runs at 20 epochs. To maintain selection flexibility, network weights were preserved at 5-epoch intervals. Following extended training sessions, the weights from Epoch 25 were ultimately established as the final model due to their optimal balance of stability and resistance to overfitting when compared to subsequent iterations. Finally, to secure dependable outcomes during user testing, Test Time Augmentation (TTA) was integrated into the evaluation pipeline [4]. By generating eight distinct transformations of the input image and averaging their resulting probabilities, TTA effectively insulated the final diagnostic outputs from lighting and angle inconsistencies.

4.3 Explainability

To bridge the "black box" gap in clinical AI, the framework incorporates Gradient-weighted Class Activation Mapping (Grad-CAM), which generates heatmaps highlighting the specific regions, such as atypical pigment networks or asymmetric borders, that drove the model's prediction. Beyond visual evidence, the system features a Personalized Safety Engine that integrates NLP-based clinical note analysis and Reinforcement Learning. Using a Linear Bandit Agent, the system adaptively tunes the diagnostic threshold (or lambda) based on patient metadata (age, sex, and anatomical site) and NLP risk scores (using DistilBERT). Furthermore, uncertainty estimation is validated through Monte Carlo (MC) Dropout in the evaluation suite, allowing the system to flag low-confidence predictions. This multi-modal approach ensures the model provides both a visual justification and a context-aware safety override to prevent diagnostic errors.

5. Evaluation

5.1 Experimental Setup

The datasets required in the model training are first downloaded and retrieved using Kaggle's API. The downloading pipeline retrieves the complete HAM10000 and ISIC 2019 dataset. This process, which includes downloading datasets and requirements, takes approximately 10-15 minutes. After the datasets have been successfully downloaded and uploaded within the training environment, which in our development we used Google Colab, we encoded each skin lesion diagnosis class with 0 (benign) and 1 (malignant). We then used these labels to classify and sort them into their respective classes, suitable for the proposed binary classification model. The following classes are encoded as 0: nv, bkl, df, and vasc. The remaining classes, namely mel, bcc, and ak/akiec, are encoded as 1. The classification algorithm used the encoded values as a basis for binary class segregation and sorting.

After sorting through the data, the dataset is then split into the standard 80% for training, 10% for validation, and 10% for testing, which will only be used after the completion of model training. In the algorithm, we included a fixed random seed of 42, so in every training session, the distribution of the data among the subsets remains the same throughout. Additionally, we implemented stratification to ensure that the subset class ratio is similar to the original class ratio.

Once all the environment and dataset configurations have been completed, the data is preprocessed using standard image formatting and Albumentations-based data augmentations. All images are resized, converted, transformed, and normalized. Albumentations techniques include robust spatial techniques such as horizontal/vertical flipping and random rotations, distortion techniques such as grid and optical distortions, and noise distortion such as Gaussian blur and color jitters (e.g, brightness, contrast, saturation, and hue).

After the training data has been properly formatted and preprocessed, the training process of the model can begin. In the original training that produced the stated 96.62% accuracy, an AU-ROC of 0.9946, a Recall of 0.9642, and a Precision of 0.9773, we ran the training pipeline for a max of 30 epochs. However, to meet the stated requirements of having a training runtime of less than or equal to 90 minutes, we limited the training epochs to 20.

Finally, to guarantee that we get the model with the best performance across all metrics, and have flexibility in selecting the final model weight, we periodically saved the model weights every 5th epoch (e.g, 5, 10, 15, 20, 25, 30).

5.2 Ablations

To verify which component used in the training process of the proposed model had the largest influence on model performance, we identified two likely components to perform ablations on. In our first ablation process, we reverted the preprocessing pipeline from the new Albumentations pipeline to the standard data augmentation techniques using PyTorch. However, this keeps and uses the balanced custom dataset. Preprocessing techniques used in this ablation are limited to image resizing, transformation, and normalization.

After running the adjusted training pipeline for 30 epochs using the same EfficientNet-B4 architecture, the ablation model, labeled as ablation_v1, achieved a performance metric of:

Metric	**Scores**
Accuracy	89.67%
F1-Score	0.9042
Recall (Sens)	0.8407
Precision	0.9780
AU-ROC	0.9805
MC-Dropout	0.004420
Sensitivity at Fixed Capacity (10% FPR)	0.9451

In our second ablation process, we removed the ISIC 2019 malignant classes from the custom dataset. This essentially limits the learning pipeline to using only the HAM10000 dataset and addresses class imbalance with usual class balancing techniques, namely the weighted random sampler and weight decay. However, this ablation uses and keeps the new Albumentation pipeline.

After running the adjusted training pipeline for 30 epochs using the same EfficientNet-B4 architecture, the ablation model, labeled as ablation_v2, achieved a performance metric of:

Metric	**Scores**
Accuracy	90.71%
F1-Score	0.7520
Recall (Sens)	0.7231
Precision	0.7833
AU-ROC	0.9527
MC-Dropout	0.003218
Sensitivity at Fixed Capacity (10% FPR)	0.8667

After analyzing the results of the two ablation processes, we can conclude that the component that has the most influence on the performance of the model is the integration of the ISIC 2019 malignant classes into the custom dataset. The results emphasized the need for utilizing a robust and balanced dataset in order for the model to accurately learn key visual features without sacrificing performance. Additionally, it proved that maximizing publicly available datasets in creating balanced datasets suitable for project objectives is essential in developing a high-performance model.

5.3 Final Model Results

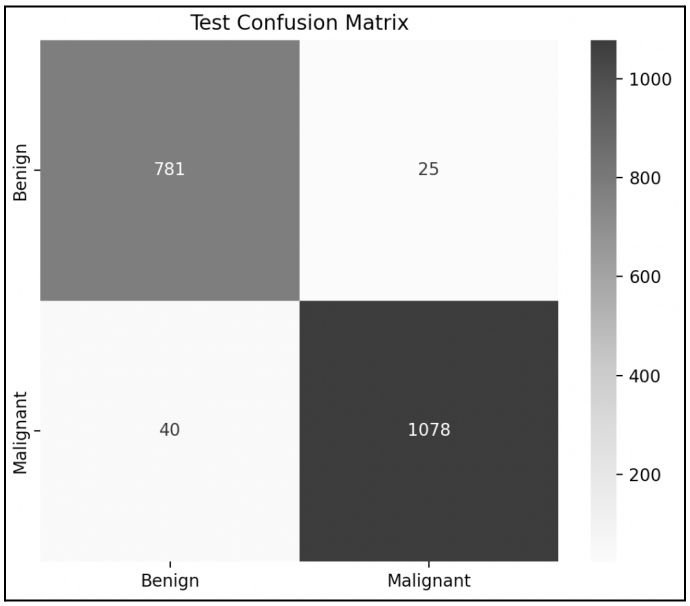
The final model, labelled as efficientnet_v2, was produced after utilizing the custom dataset and Albumentations-based pipeline that ran for a maximum of 30 training epochs. To conclude the training process and finally identify the actual performance metrics and capabilities of the chosen model, we ran the final check using the test subset. The efficientnet_v2 model achieved performance metrics of:

Metric	**Scores**
Accuracy	96.62%
F1-Score	0.9707
Recall (Sens)	0.9642
Precision	0.9773
AU-ROC	0.9946
MC-Dropout	0.000550
Sensitivity at Fixed Capacity (10% FPR)	0.9893

Based on the achieved metrics, the proposed model has demonstrated exceptional performance and high confidence in accurately classifying skin lesions as either malignant or benign, as well as minimizing false positives and false negatives. The model's high accuracy predicts the

right diagnosis for the skin lesion 96.62% of the time, and the model's classification is further supported by its high precision and recall. The AU-ROC further proves the model's ability to distinguish benign and malignant visual patterns, which further proves that the learning framework and pipeline used in the training process are effective for developing high-performing models.

The model's error-balancing capabilities are proven by its high precision and recall. Whenever the model predicts a positive diagnosis, it is 97.73% accurate most of the time, which indicates a low possibility of false positives. Conversely, the model successfully identifies 96.42% of all actual positive diagnoses. Additionally, the sensitivity of 0.9893 demonstrates that the model is extremely sensitive to identified key features, meaning it misses a near-zero actual positive diagnosis, indicating a low false negative rate. The MC-Dropout demonstrates the model's confidence in its classifications, meaning that the model is both stable and effective in accurate predictions.



The confusion matrix outlines the model's effectiveness and robust performance during the final testing process. It achieved high accuracy and correctly identified 1,078 malignant and 781 benign skin lesions. However, the model misclassified 40 malignant lesions as benign, and misclassified 25 benign lesions as malignant. While there are 40 false negatives identified by the model, the model complements it through its precision of 0.9773 and recall of 0.9642, which are both balanced and high. Furthermore, the sensitivity at fixed capacity boasts a high value of 98.93%, meaning it can catch a majority of skin lesion cancers and can be further fine-tuned to prioritize patient safety.

6. Conclusion

The primary objective of the project was to develop a skin lesion classification model that prioritizes clinical safety over perfect accuracy, which was fully achieved at the end of the development phase. The project successfully produced a final, baseline model that remains efficient under various types of distortions. By incorporating explainability (Grad-CAM) and uncertainty estimates (MC Dropout), the project met the MVP requirements for a transparent system that not only classifies but also provides visual evidence for its decision or prediction.

The most significant insight derived from the design, development, and testing of the model was that the dataset parity should be the priority in medical machine learning. While the original HAM10000 dataset was large and, in terms of volume, sufficient, its 80% benign bias acted as a statistical noise or bias source that prevented the classification model from learning malignant features. The integration of the ISIC 2019 Challenge dataset and the choice of using the malignant samples proved to be the turning point and best choice for improvement, which can be seen through the versions of the model prior to ISIC 2019. The dataset provided the necessary amount of images to achieve balance and for the EfficientNet-B4 architecture to better identify skin lesion markers or characteristics like pigment and asymmetric lesion borders. Furthermore, the use of Albumentation confirmed that models trained on imperfect data are more robust when faced with real-world, user-captured images that suffer from distortion such as lens warping, color distortion or difference, and poor lighting.

The primary performance metrics at Epoch 25 suggest that the classification model is not only identifying almost all instances of skin lesions, but is doing so with very few false alarms, as previous and later epoch versions have predicted incorrect or false classifications. The metrics, such as its 96.68% Recall, 97.46% Precision, and 0.9702 F1-Score, serve as proof that the model achieved a good balance between sensitivity and reliability. Most importantly, the reduction of the error rate from 33% of previous versions to 3.4% of the latest, final baseline model proves that the implementation of dataset parity and Albumentation positively improves the robustness and performance of the classification model.

7. Ethics & Policy

A critical ethical consideration in the deployment of this skin lesion classification system revolves around demographic representation and algorithmic bias. The primary dataset utilized for training, HAM10000, predominantly features lighter skin tones sourced from populations in Austria and Australia. This lack of diversity introduces a significant risk of skewed model performance, where the system may demonstrate decreased accuracy when analyzing lesions on underrepresented or darker skin types. Such disparities prompt vital questions regarding the equity, broader clinical usability, and reliability of the diagnostic tool across diverse patient populations.

Furthermore, the statistical distribution of the initial dataset posed a profound clinical safety challenge. With approximately 80% of the cases labeled as non-cancerous, a model could artificially achieve 80% accuracy simply by classifying every instance as benign, completely overlooking harmful abnormalities. Because misclassifying a malignant growth as harmless constitutes a massive safety risk to patients, pure accuracy was deemed a distorted and insufficient metric. Consequently, the development policy shifted heavily toward optimizing sensitivity, precision, and the F1-score to rigorously limit the occurrence of missed detections and dangerous false negatives. Exactness was also carefully monitored, as frequent false alarms could cause undue clinical distress.

Another serious ethical risk emerges regarding user interaction and the interpretation of the model's outputs. There is an inherent danger that individuals might accept the algorithmic results as definitive medical verdicts, bypassing necessary professional consultations and further clinical research. To mitigate this risk, strict policies were implemented to display explicit disclaimers within the system interface. These disclaimers clearly state that the application is designed purely for academic and research purposes, is not approved for operational

clinical use, and cannot replace an actual consultation with a qualified healthcare professional.

Finally, to prevent the system from operating as an opaque "black box" and to foster user trust, robust explainability measures were integrated directly into the diagnostic framework. Gradient-weighted Class Activation Mapping (Grad-CAM) and saliency maps were employed to provide visual highlights of the specific image areas and patterns that influenced the model's decision. Additionally, Monte Carlo (MC) Dropout was utilized to generate measurable confidence estimates and uncertainty levels derived from repeated sampling trials. This dual approach ensures that users can independently verify the visual reasoning behind a prediction and are explicitly alerted to low-confidence results, encouraging them to proceed with appropriate caution.

8. Recommendation

Based on experimental results, the researchers recommend that future iterations transition toward a fully integrated multimodal architecture. While the current system utilizes a "safety lock" to combine vision and text, a deeper integration, where patient age, history, and anatomical location are injected directly into the neural network's feature layers, could further refine prediction accuracy.

Furthermore, future development should prioritize the acquisition of diverse datasets representing varied skin phototypes to eliminate ethnic bias. Comparative analysis between other model weights, such as the other EfficientNet variants and ResNet models, could further expand skin lesion classification ability. Finally, expanding the reinforcement learning component to include more clinical actions, such as recommending immediate biopsy or 6-month follow-up, would move the system toward a more comprehensive clinical decision support tool.

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Appendix A: Model Card

The skin lesion classifier utilizes an EfficientNet-B4 backbone, originally pre-trained on ImageNet and fine-tuned for high-sensitivity binary classification of benign versus malignant lesions. This model was developed using a combined dataset of 19,335 clinically curated images, sourced from the HAM10000 and ISIC 2019 repositories, specifically balanced to mitigate the high-frequency bias of benign samples found in standard dermatological sets. Training was conducted using the AdamW optimizer with a learning rate of 0.0002 and a weighted cross-entropy loss function to prioritize the detection of malignancies, achieving a final test accuracy of 96.62%. The architecture is bolstered by an integration layer that allows for context-aware thresholding through a Linear Bandit reinforcement

learning agent, which adjusts the classification boundary based on patient metadata and clinical narratives. Performance was further validated using Monte Carlo (MC) Dropout for uncertainty estimation and Grad-CAM for spatial feature localization, ensuring that the model's diagnostic logic is both robust and interpretable for clinical screening.

Appendix B: Software Availability

The complete experimental pipeline is available via a modularized Python repository designed for reproducibility. The environment setup and data acquisition are automated through the get_data.py script, which handles the secure downloading of the HAM10000 and ISIC 2019 datasets via the Kaggle API. The software architecture is organized into distinct modules: src/train.py for the EfficientNet pipeline, src/nlp_component.py for clinical text processing, and src/rl_threshold.py for the adaptive decision logic. Full dependency management is handled via a requirements.txt file, supporting both CUDA and Apple Silicon (MPS) acceleration.

B.1 Reproduction and Environment Setup

The reproduction script includes automated downloading of the HAM10000 and ISIC 2019 datasets from Kaggle, downloading dependencies based on the requirements file, as well as providing users with options to evaluate the final model (efficientnet_v2) and run the GUI-based explainability program. To access all these, simply run the “sh run.sh” command in the project root terminal. This should work in Linux, MacOS, and Windows operating systems.

The program makes use of Python virtual environments. Downloading of requirements and datasets is strictly required to ensure that no errors will arise during the execution of the program. To run the training process, simply run the train.py script located in the /src directory of the project. To only run the GUI explainability program, simply run the explainability.py script located in the /src directory of the project.

B.2 Demonstration

The project demonstration features a functional GUI-based diagnostic dashboard, which is explainability.py, that allows users to interact with the trained EfficientNet-B4 model in real-time. Users can upload a lesion image, input clinical notes (e.g., "rapidly growing and itchy"), and receive a combined diagnostic report. The dashboard displays the original image alongside the Grad-CAM heatmap, the vision-based probability, and the context-adjusted safety threshold. This demo showcases the model's robustness under simulated distortions; it highlights how the NLP-driven "Safety Lock" overrides vision confidence when high-risk clinical markers are detected, maintaining a high test accuracy of 96.62%, 96.68% Recall, 97.46% Precision, and 0.9702 F1-Score.